Executive Summary — Surrogate Model Validation Report

Use case: UCHARDLANDING_1 | 09 September 2026 | 2 model(s)

ALL REQUIREMENTS MET

Recommended model: GradientBoosting

Avg R^2: 0.9836 — Excellent

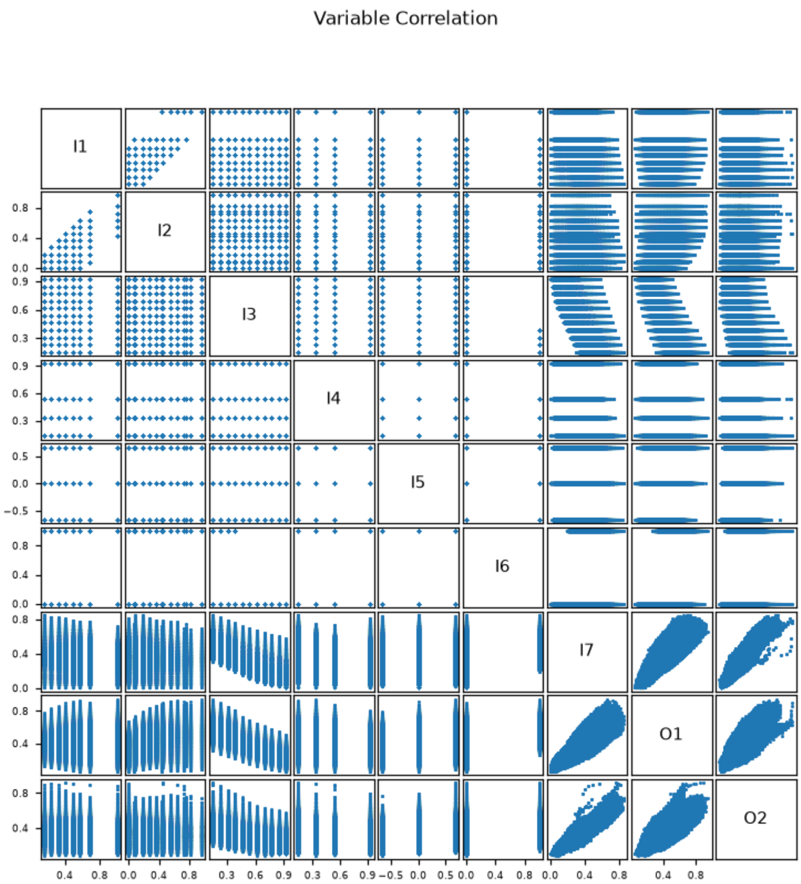
Q90 Accuracy — All outputs pass

Pass: O1, O2

Project Overview

Use case	UCHARDLANDING_1
Inputs	I1, I2, I3, I4, I5, I6, I7
Outputs	2: O1, O2
Train / Val / Test	80% / 10% / 10%
Accuracy target	Q90 < 10% relative error per output

Variable Correlation — Input vs Output



Data Statistics

Per-variable statistics over all data (train + val + test), as `Train_set.describe()` reports them in the feature-selection notebook: count, mean, standard deviation, minimum, quartiles and maximum.

Inputs

	count	mean	std	min	P25	P50	P75	max
I1	89,357	0.5276	0.2292	0.1646	0.3292	0.4938	0.679	1
I2	89,357	0.3137	0.2394	0	0.0909	0.2727	0.4545	0.9636
I3	89,357	0.4757	0.2439	0.1538	0.2308	0.3846	0.6923	0.9231
I4	89,357	0.4694	0.2875	0.1333	0.1333	0.3333	0.5333	0.93
I5	89,357	-0.000187	0.5404	-0.6667	-0.6667	0	0.6667	0.6667
I6	89,357	0.2577	0.4374	0	0	0	1	1
I7	89,357	0.3477	0.1585	0.0042	0.2366	0.3448	0.4558	0.8502

Input variables — Train_set.describe()

Outputs

	count	mean	std	min	P25	P50	P75	max
O1	89,357	0.4207	0.154	0.0511	0.3061	0.4217	0.5289	0.9489
O2	89,357	0.3301	0.1204	0.0833	0.2367	0.326	0.411	0.9167

Output variables — Train_set.describe()

Model Selection

Model	Avg R^2	Q90 pass (2)	KS pass (2)
GradientBoosting ⋆	0.9836	2/2	2/2
MLP	0.9828	2/2	2/2

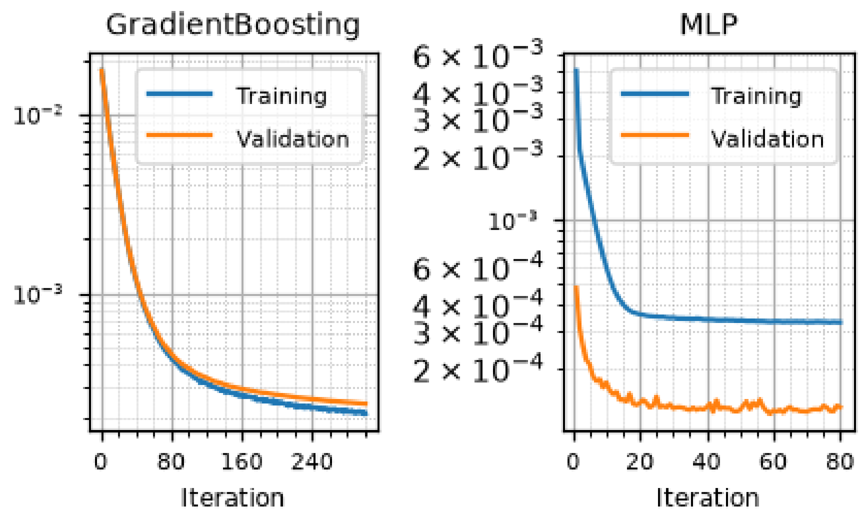
⋆ Recommended model. Q90 target: $< 10\%$. KS: $p \geq 0.05$ = no overfitting.

Accuracy Assessment (Q90 & R^2)

Output	GradientBoosting ⋆			MLP		
	R^2	Q90	Gap	R^2	Q90	Gap
O1	0.9960	0.0407	-0.0593	0.9952	0.0443	-0.0557
O2	0.9712	0.0870	-0.0130	0.9705	0.0851	-0.0149

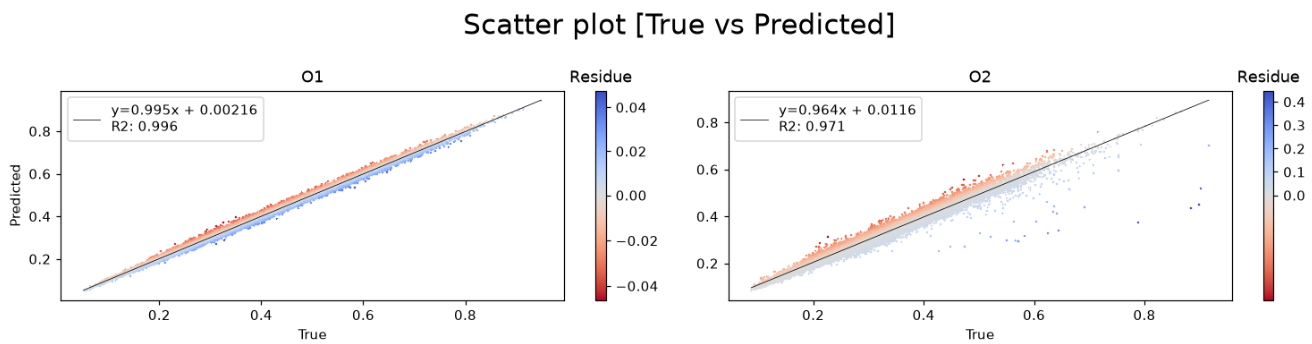
$R^2 \geq 0.95$ $R^2 \in [0.80, 0.95)$ $R^2 < 0.80$ or Q90 fail pass Gap=Q90−target ($<0 \Rightarrow$ margin)

Training History



Training and validation loss per iteration, log scale. The validation curve is measured, not approximated: per epoch on the early-stopping split for the MLP, per stage on the validation split for gradient boosting (whose training curve is its per-stage deviance). Models saved before this was recorded fall back to the $(1 - R^2) \cdot \mathrm{Var}(y)/2$ approximation.

Predicted vs True — GradientBoosting



Data Split Quality

Metric	Value	Pass
Residual voxel proportion ($\leq$0.05)	0.000	✓
Valid test proportion	0.968	
p-hacking test proportion ($\leq$0.05 warn)	0.009	✓
Isolated test proportion ($\leq$0.05 warn)	0.024	✓
Chi^2 p-value ($\geq$0.05)	—	—

No p-hacking concern: only 0.9% of test points lie closer to a training point than training points are to each other, so the test error is not flattered.

Overfitting Check (KS Test) — GradientBoosting

Output	KS p-value
O1	0.074
O2	0.162

$p \geq 0.05$ — no overfitting $p < 0.05$ — overfitting detected

Deep Analysis

UCHARDLANDING_1

Winner Model — Full Validation

Deep Analysis — GradientBoosting

Mirrors `validation_template.ipynb` — all computations from validation CSVs

Data Overview

Statistical summary of inputs and outputs over all data (train + val + test), as reported by `describe()` in the feature-selection notebook: count, mean, standard deviation, minimum, quartiles and maximum per variable. All figures in this part are drawn by `validationlib`, matching the extended validation notebook.

Input Statistics

	count	mean	std	min	P25	P50	P75	max
I1	89,357	0.5276	0.2292	0.1646	0.3292	0.4938	0.679	1
I2	89,357	0.3137	0.2394	0	0.0909	0.2727	0.4545	0.9636
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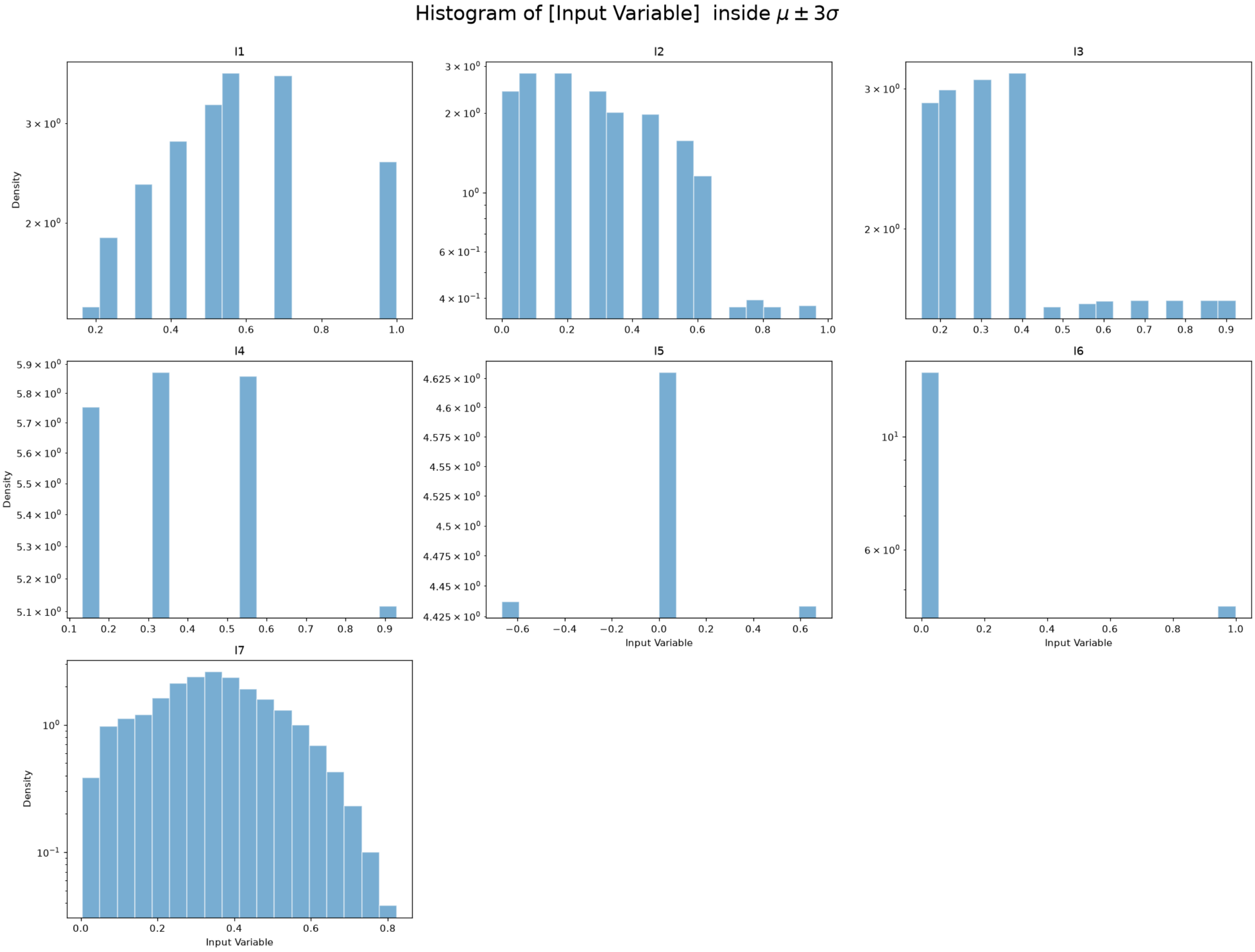
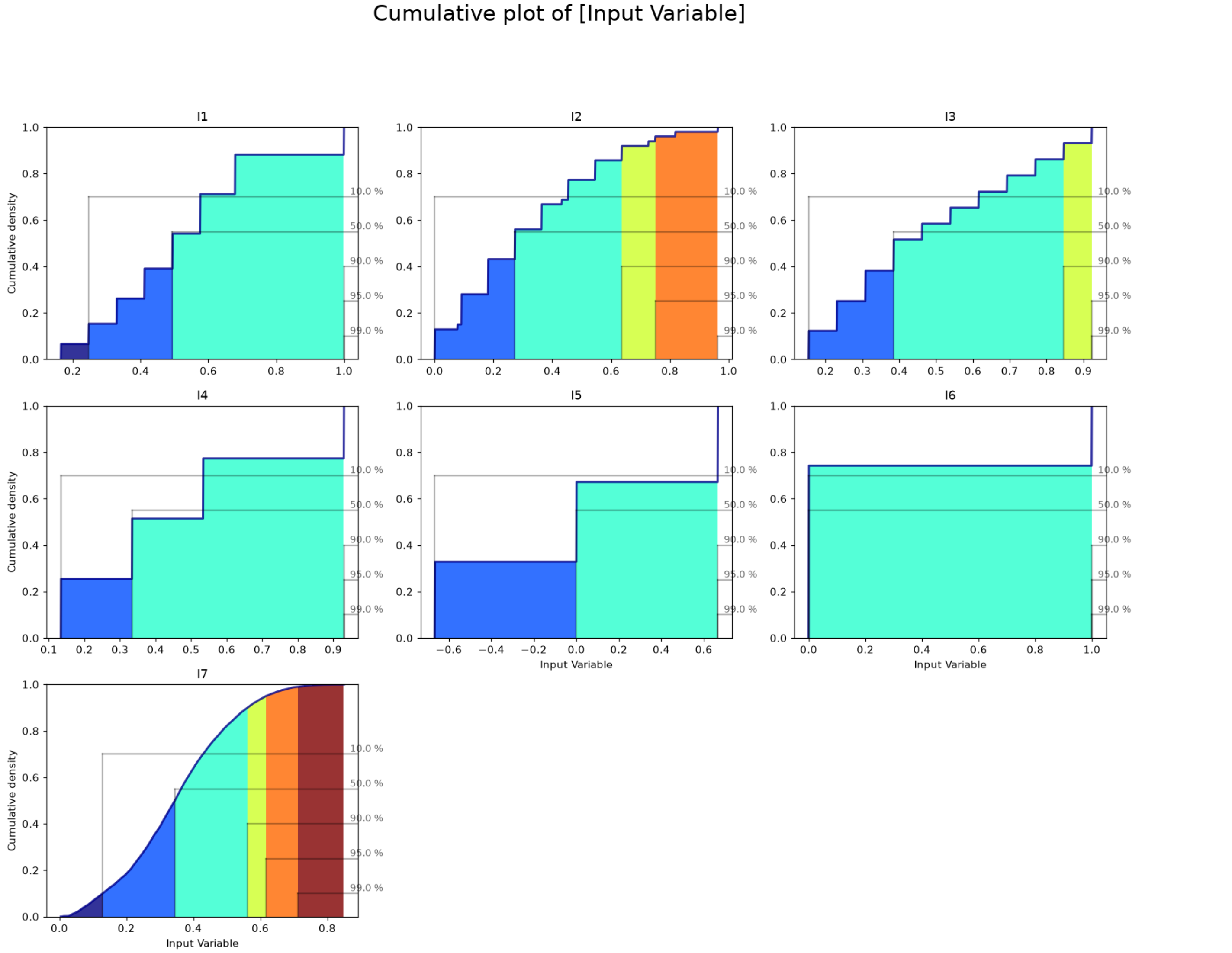
I5	89,357	-0.000187	0.5404	-0.6667	-0.6667	0	0.6667	0.6667
I6	89,357	0.2577	0.4374	0	0	0	1	1
I7	89,357	0.3477	0.1585	0.0042	0.2366	0.3448	0.4558	0.8502

Input variables — Train_set.describe()

Input Histograms



Input Cumulative Distributions

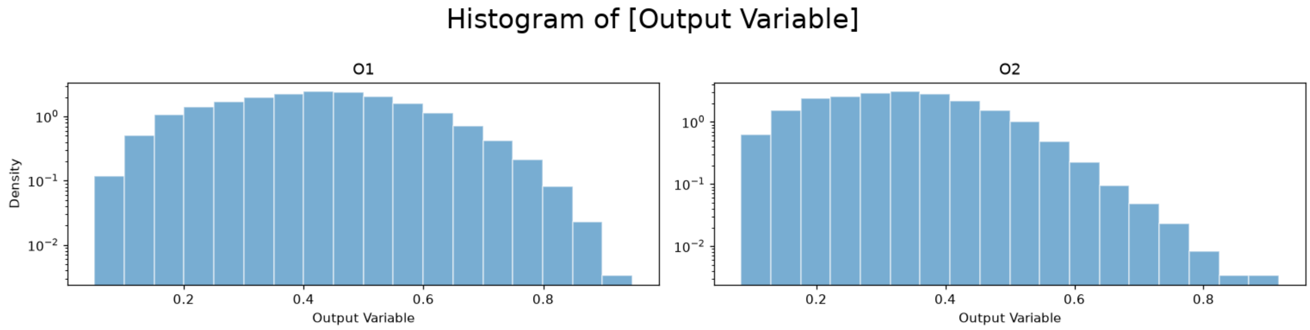


Output Statistics

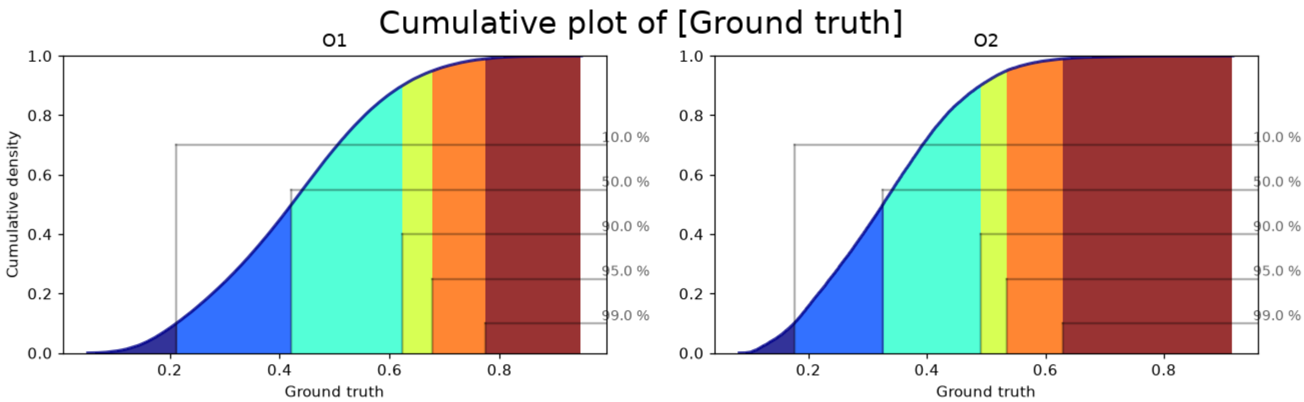
	count	mean	std	min	P25	P50	P75	max
O1	89,357	0.4207	0.154	0.0511	0.3061	0.4217	0.5289	0.9489
O2	89,357	0.3301	0.1204	0.0833	0.2367	0.326	0.411	0.9167

Output variables — Train_set.describe()

Output Histograms



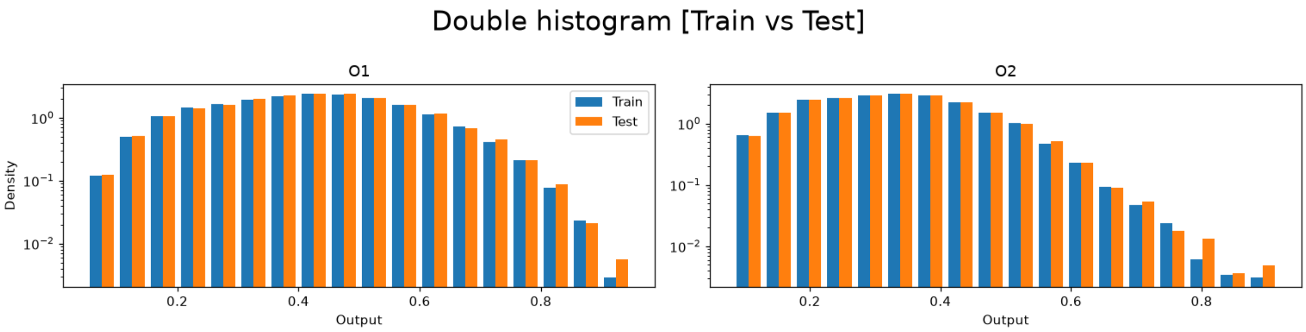
Output Cumulative Distributions



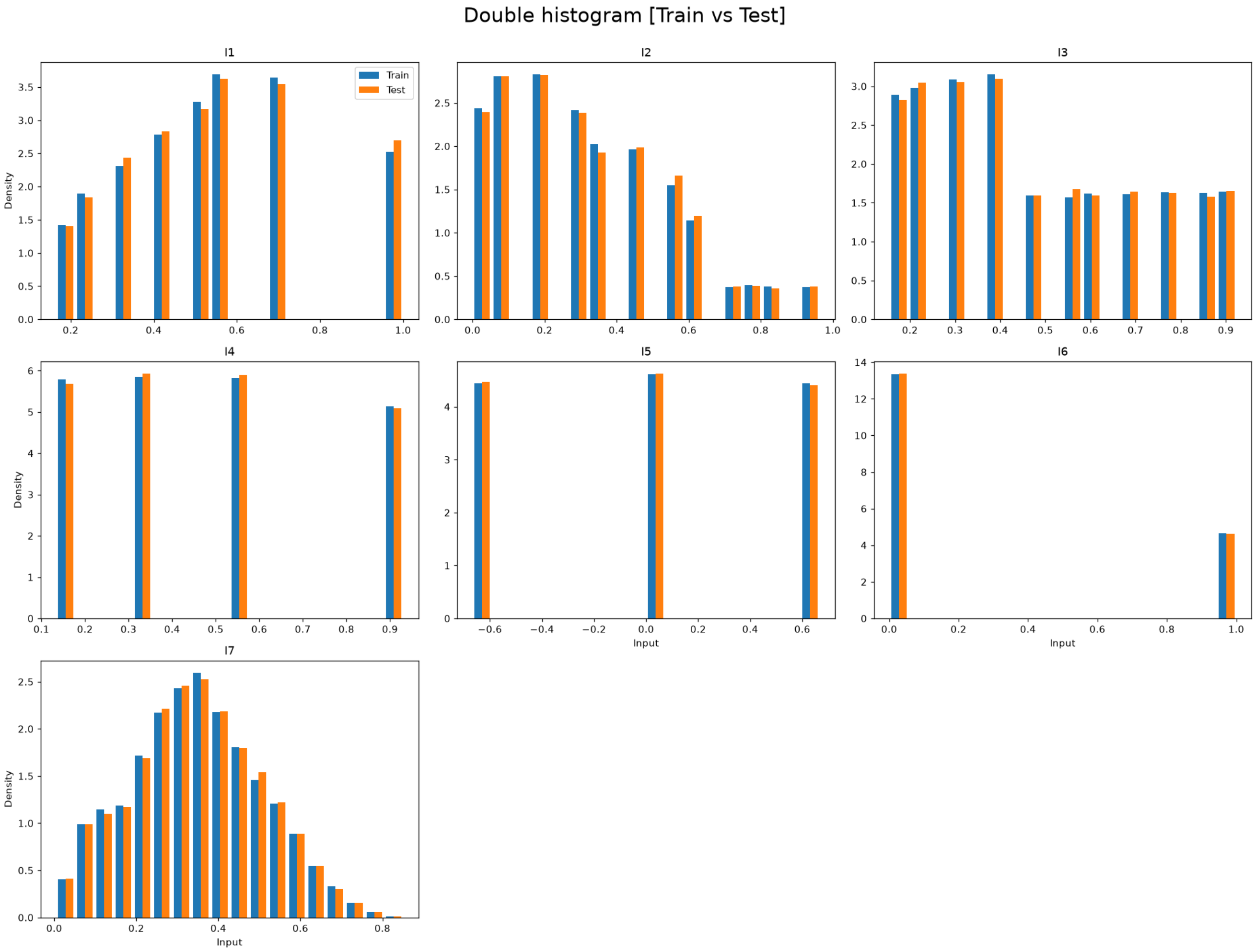
Train-Test Split Analysis

A well-designed split produces training and test distributions that are similar but not identical. The Kolmogorov-Smirnov (KS) test and the Anderson-Darling (AD) test are used to assess distributional similarity. H_0: both sets come from the same distribution. $p < 0.05$ rejects H_0 (undesirable if too low; undesirable if the split was done by stratification and p is too high).

Output Distributions: Train vs Test



Input Distributions: Train vs Test



KS and AD Test Results

	KS	AD
O1	0.684	0.25
O2	0.6428	0.25

Train vs test p-values. Red: $p < 0.05$ (distributions differ).

Split Quality — Voxel Tesselation Proximity (p-hacking check)

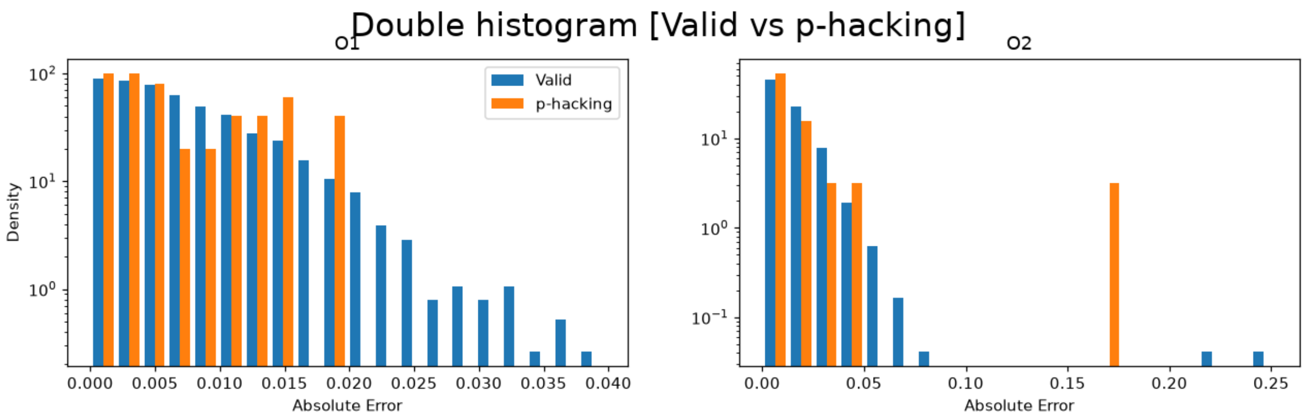
Beyond matching distributions, a split can still flatter the model if test points sit right next to training points (p-hacking) or probe regions the model never saw (isolated, residual voxel). The VTPM method from `validationlib` (SF_9 cell 9.0) classifies every test point.

Metric	Value	Pass
Residual voxel proportion (≤0.05)	0.000	✓
Valid test proportion	0.968	
p-hacking test proportion (≤0.05 warn)	0.009	✓
Isolated test proportion (≤0.05 warn)	0.024	✓
Chi^2 p-value (≥0.05)	—	—

No p-hacking concern: only 0.9% of test points lie closer to a training point than training points are to each other, so the test error is not flattered.

Error Distribution: Valid vs p-hacking Test Points

Absolute error of the test points flagged as p-hacking against the valid ones, exactly as the extended validation notebook draws it. Matching distributions (high p-values) mean the flagged points do not distort the reported error.

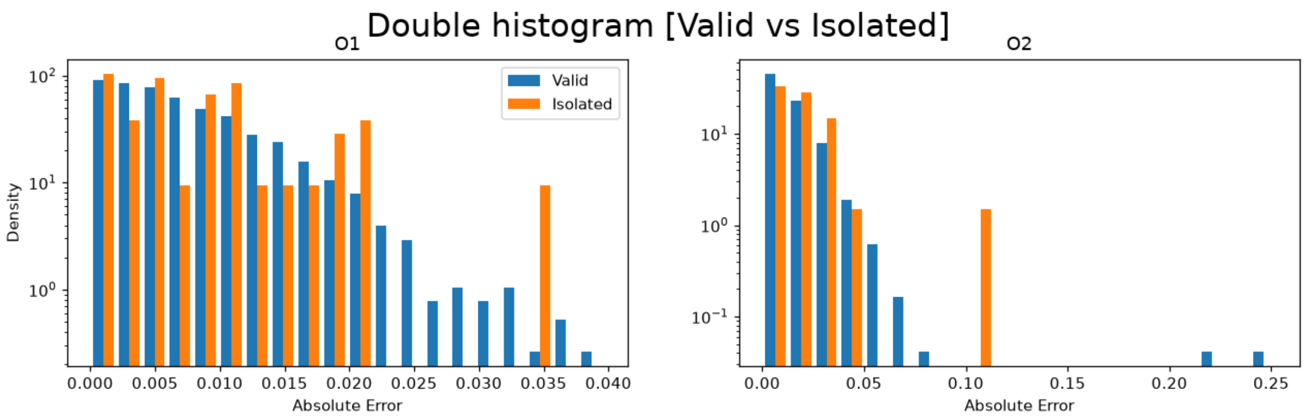


	AD	KS	KW
O1	0.25	0.7073	0.9937
O2	0.25	0.085	0.2566

Valid ($n = 1922$) vs p-hacking ($n = 25$) absolute-error distributions, on a subsample of the test set. Red: $p < 0.05$ (the flagged points behave differently).

Error Distribution: Valid vs Isolated Test Points

Same comparison for the test points with no training neighbour.



	AD	KS	KW
O1	0.2335	0.143	0.2572
O2	0.1288	0.1172	0.1187

Valid ($n = 1922$) vs Isolated ($n = 53$) absolute-error distributions, on a subsample of the test set. Red: $p < 0.05$ (the flagged points behave differently).

Error Quantification — P(E)

Two error metrics are studied:

- Residue: $e_i = y_i - \hat{y}_i$. A centred ($\mu \approx 0$), symmetric distribution indicates an unbiased model.
- Absolute Error: $|e_i|$. Always non-negative; its Q90 must satisfy the accuracy requirement.

Residue Statistics Table

Bootstrap confidence bounds shown as superscript (upper) and subscript (lower).

	count	min	max	mean	(BS)mean	median	(BS)median	std	(BS)std	IQR	(BS)IQR	kurtosis	(BS)kurtosis	skewness	(BS)skewness
O1	17,872	-0.0462	0.0476	4.66e-05	$^{0.000184}_{-9.85e-05}$	9.86e-05	$^{0.000231}_{-6.67e-05}$	0.0098	$^{0.0099}_{0.0096}$	0.012	$^{0.0121}_{0.0117}$	0.7	$^{0.823}_{0.59}$	-0.0602	$^{-0.0148}_{-0.118}$
O2	17,872	-0.0848	0.449	0.000286	$^{0.000586}_{-3.35e-05}$	-0.000407	$^{-0.000115}_{-0.000645}$	0.0205	$^{0.0215}_{0.0193}$	0.0209	$^{0.0213}_{0.0205}$	63.8	$^{89.1}_{31.2}$	4.09	$^{5.22}_{2.43}$

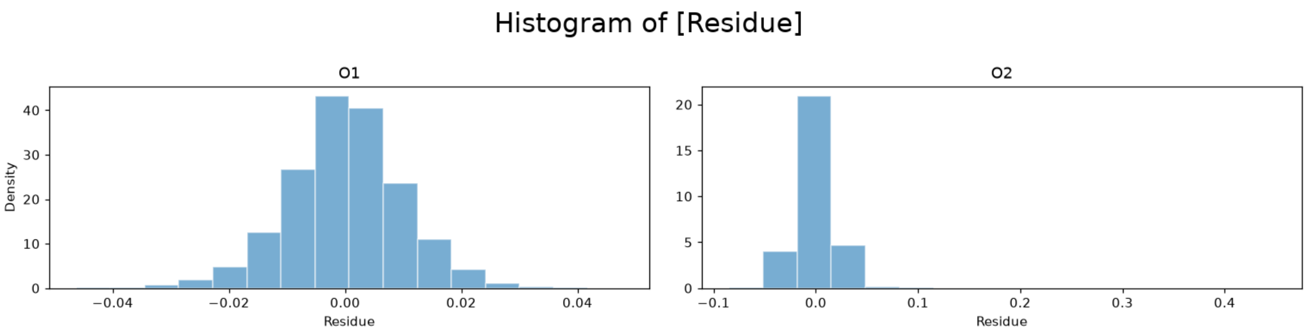
Residue statistics with bootstrap confidence intervals (superscript/subscript bounds).

Absolute Error Statistics Table

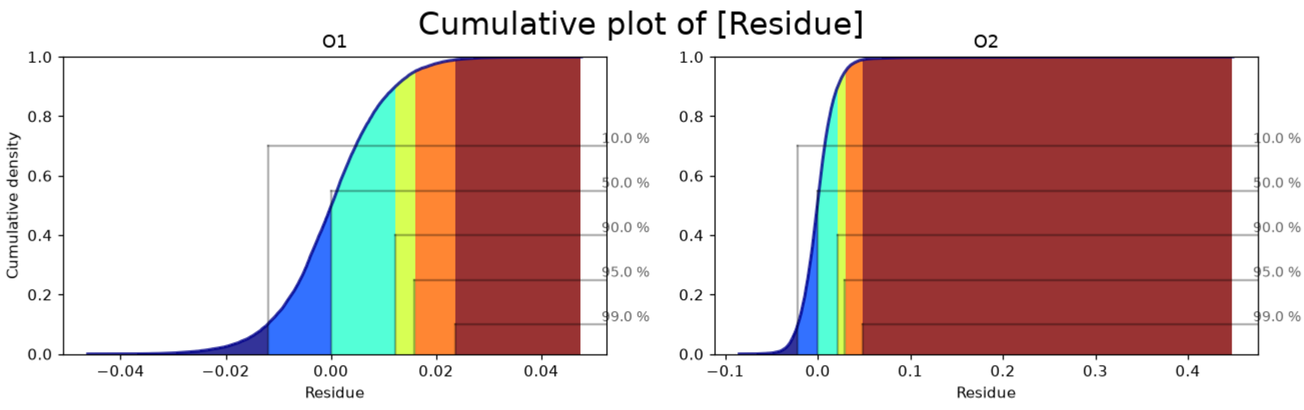
	count	min	max	mean	(BS)mean	median	(BS)median	std	(BS)std	IQR	(BS)IQR	kurtosis	(BS)kurtosis	skewness	(BS)skewness
O1	17,872	1.32e-07	0.0476	0.0075	$^{0.0076}_{0.0075}$	0.006	$^{0.0061}_{0.0059}$	0.0062	$^{0.0063}_{0.0061}$	0.008	$^{0.0081}_{0.0078}$	2.18	$^{2.53}_{1.85}$	1.33	$^{1.39}_{1.27}$
O2	17,872	1.87e-07	0.449	0.0137	$^{0.0139}_{0.0135}$	0.0105	$^{0.0106}_{0.0103}$	0.0152	$^{0.0167}_{0.0138}$	0.0142	$^{0.0146}_{0.0139}$	178	$^{242}_{106}$	8.98	$^{11.1}_{6.6}$

Absolute-error statistics with bootstrap confidence intervals.

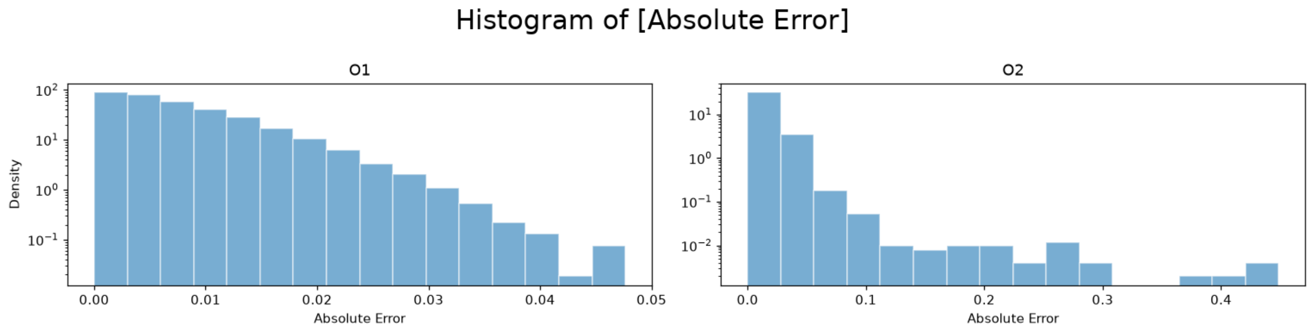
Residue Histograms



Residue CDF



Absolute Error Histograms



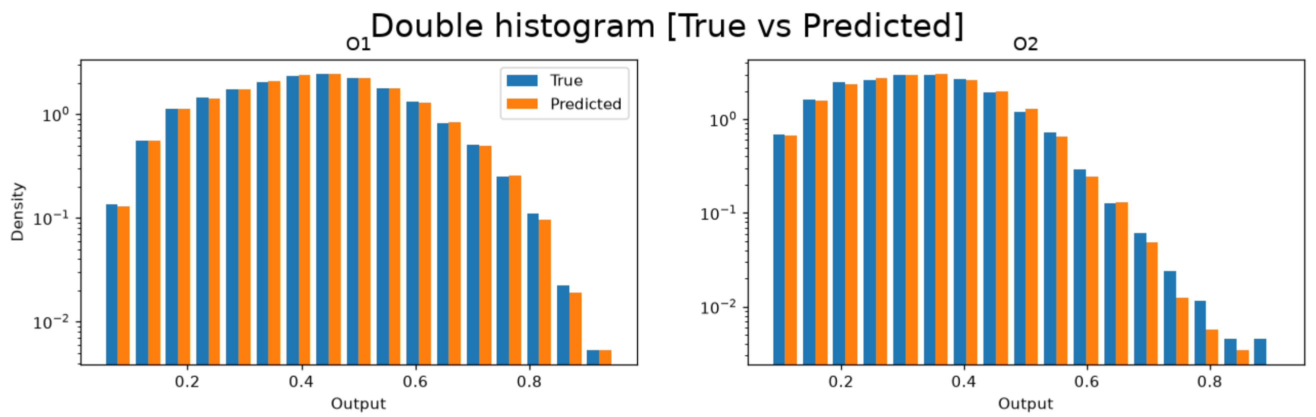
Q90 Accuracy Requirement

Output	Q90	Target	Status
O1	0.0422	10%	PASS
O2	0.0904	10%	PASS

$p \geq 0.05$ — pass $p < 0.05$ — fail

True vs Predicted Distribution Comparison

AD and KS tests comparing the distribution of true vs. predicted values on the test set. A significant difference indicates the model is not reproducing the output distribution faithfully.

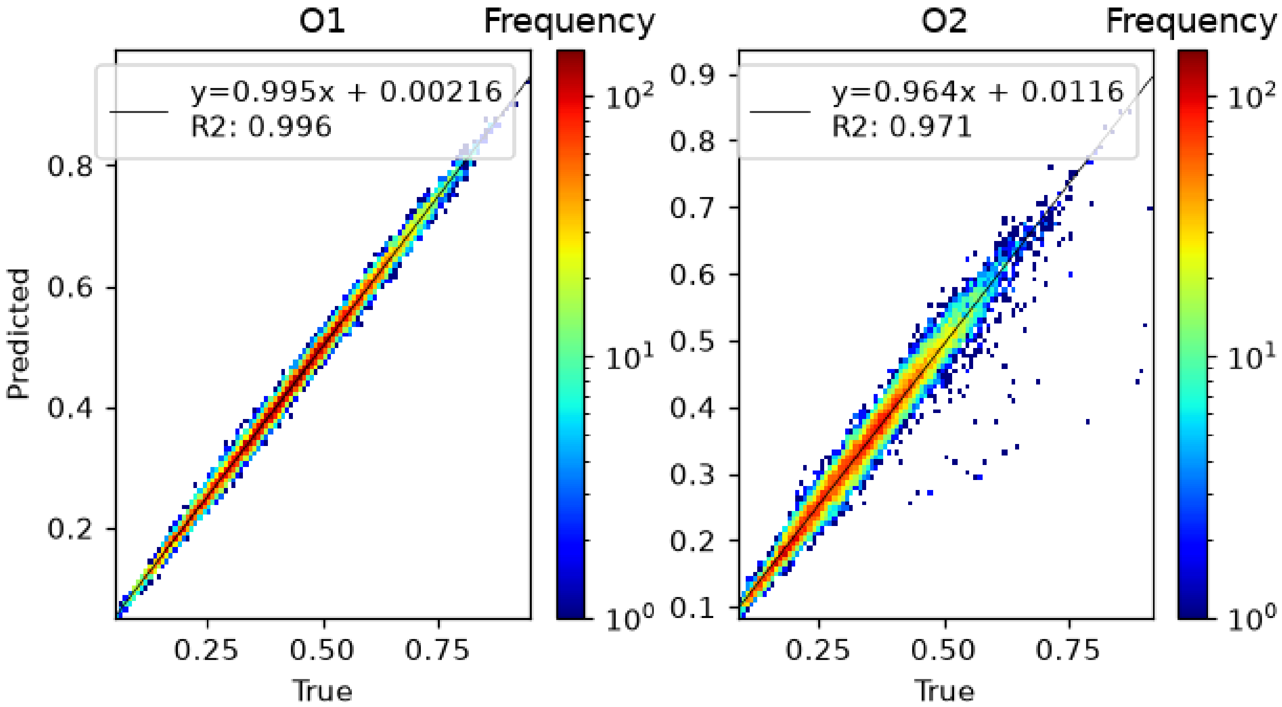


	AD	KS
O1	0.25	0.9982
O2	0.25	0.4846

True vs predicted distribution tests. Red: $p < 0.05$.

True vs Predicted — 2D Histogram

Log-scaled frequency with a linear fit; the normalized slope should be close to 1.



P(E|X) — Error Conditional on Inputs

We study whether the model error depends on the input variables. Two approaches are used: (1) a Kruskal-Wallis (KW) test on binned inputs (Sturges rule for bin count), and (2) violin plots showing the error distribution within each input bin.

Significant input-conditional bias ($p < 0.05$) detected for: I1→O1, I2→O1, I3→O1, I4→O2, I5→O1, I5→O2, I7→O1, I7→O2. These inputs explain residual variance — consider polynomial features or interaction terms.

Bias Detection Table (KW p-values)

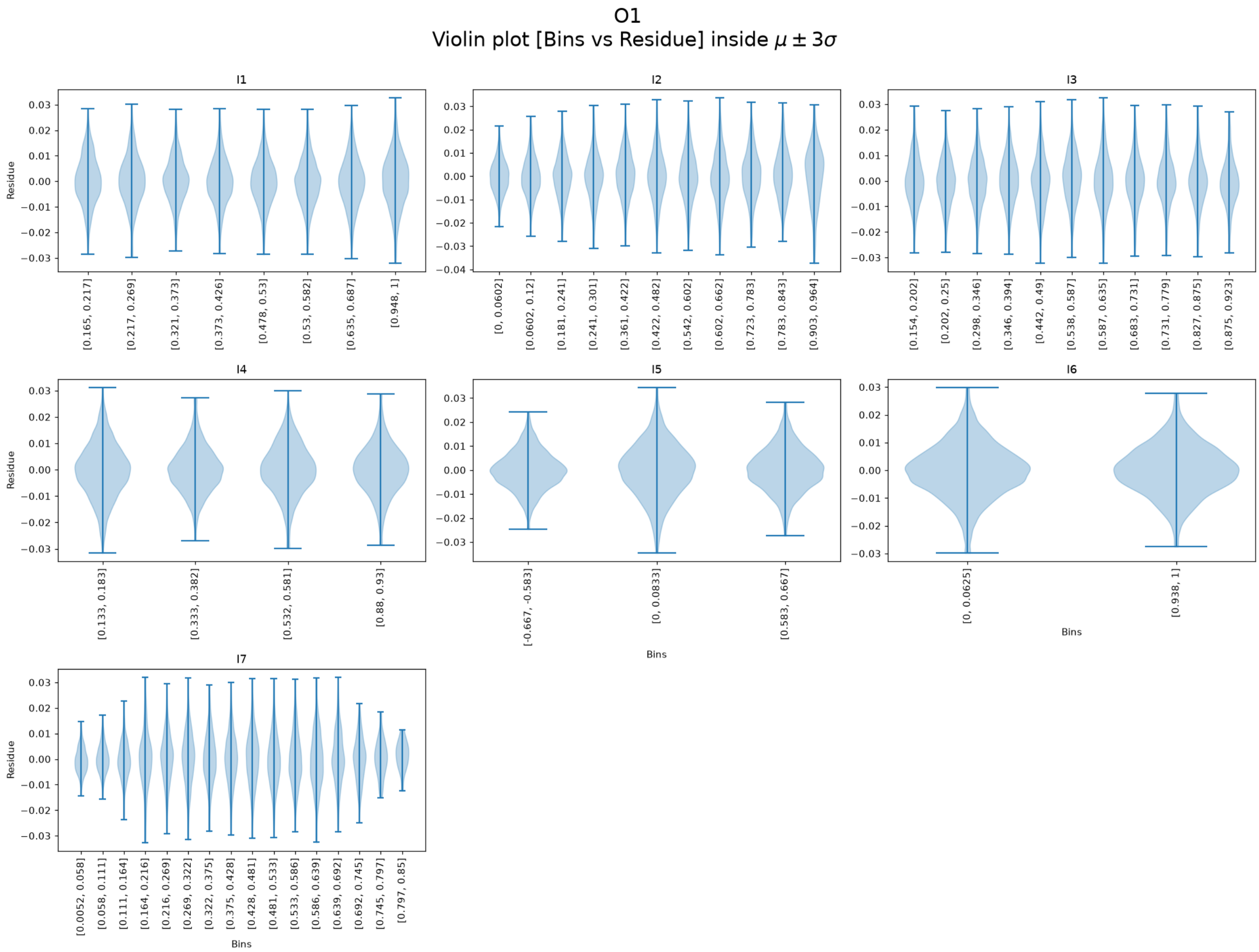
Green ($p \geq 0.05$) = residue uniform across input bins; red ($p < 0.05$) = significant bias.

	O1	O2
I1_binned	0.0044	0.1983
I2_binned	0.0421	0.4585
I3_binned	9.9e-05	0.0599
I4_binned	0.1366	0.0342
I5_binned	0.0039	2.44e-05
I6_binned	0.5181	0.0598
I7_binned	0.0251	0.0027

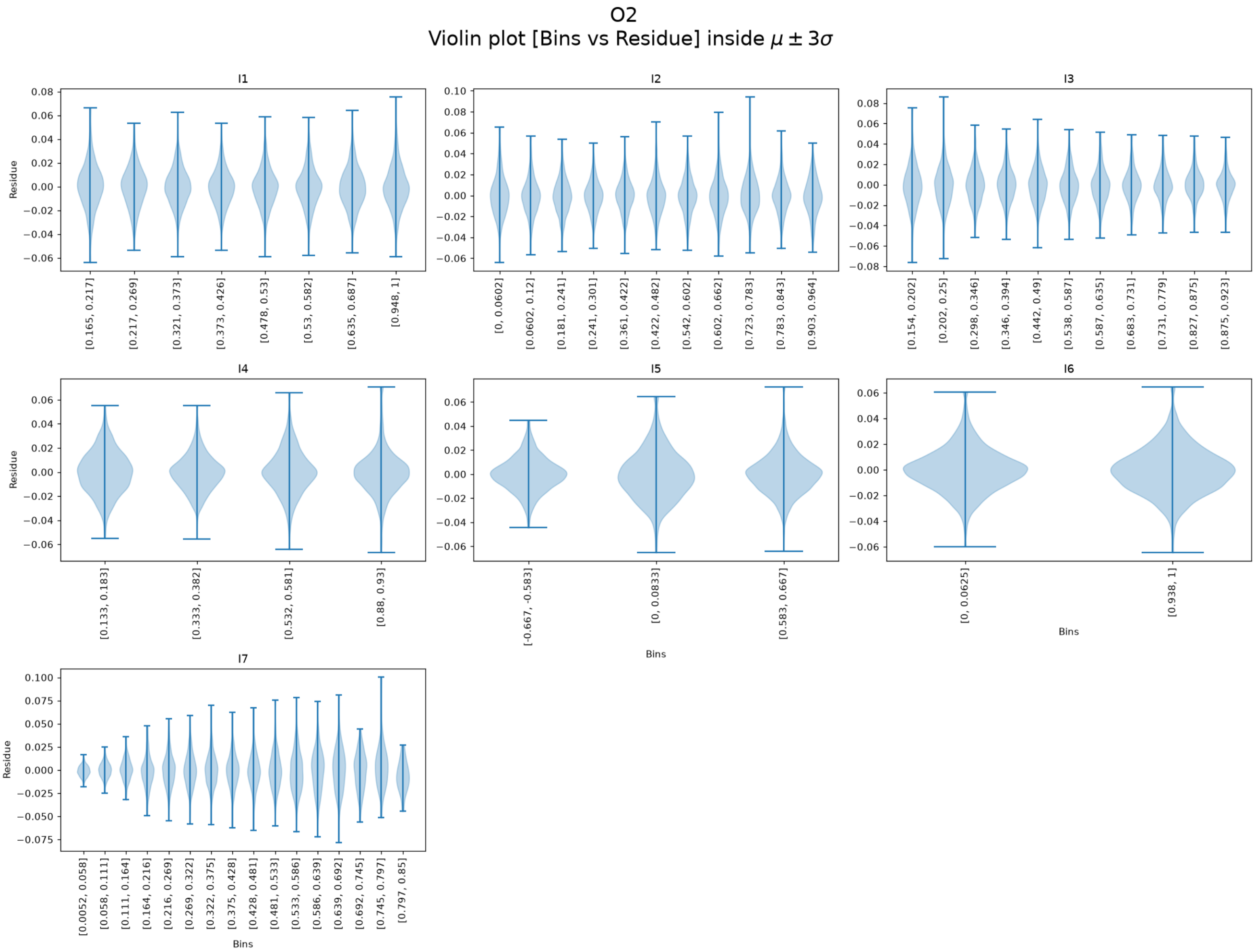
Kruskal-Wallis p-values on Sturges-binned inputs. Red: $p < 0.05$ (error depends on that input).

Residue Violin Plots per Output (binned by input)

Each panel shows the residue distribution for each input bin. A flat median line (dashed at 0) and symmetric violins indicate absence of bias.
Residue — O1

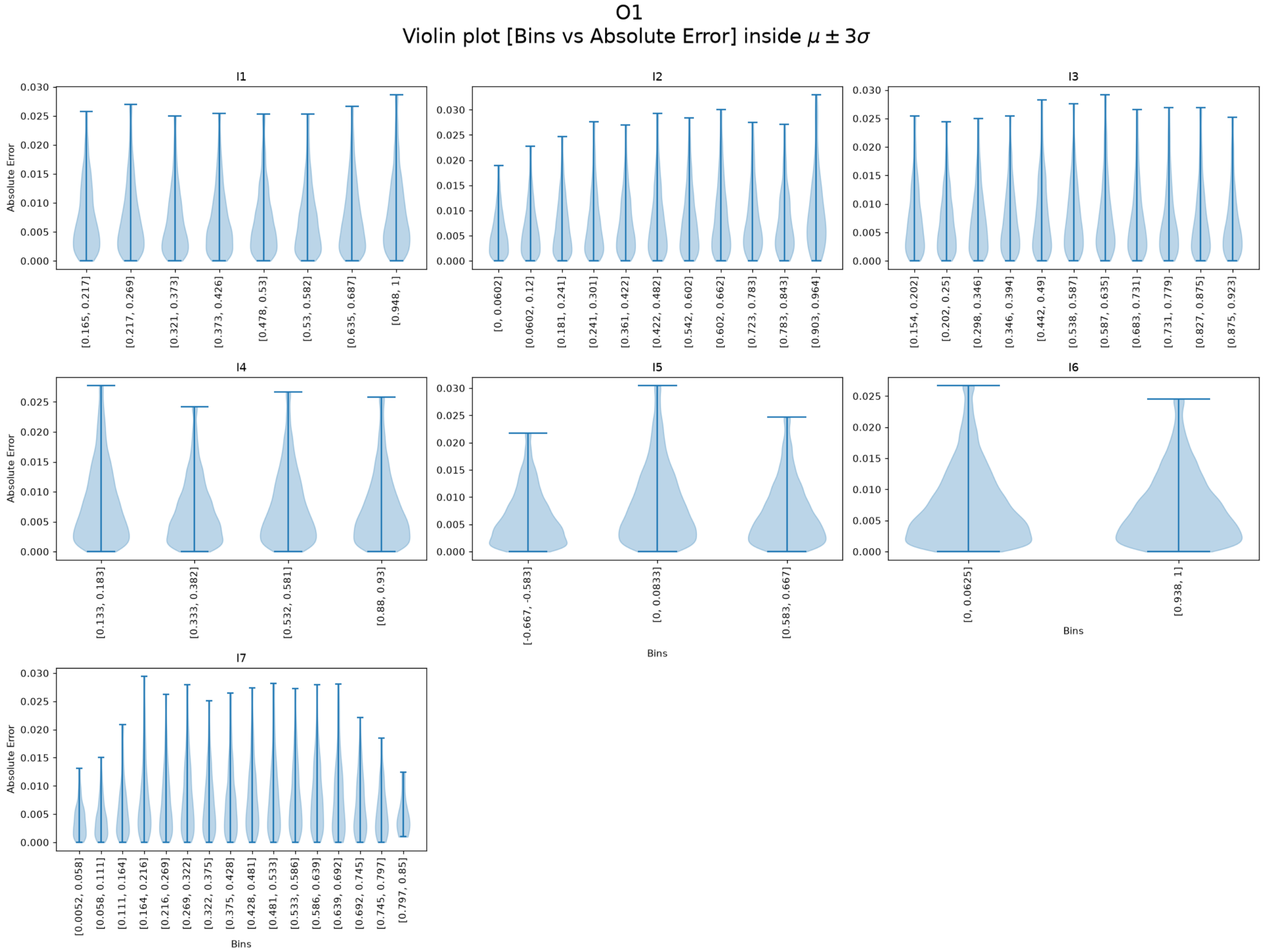


Residue — O2

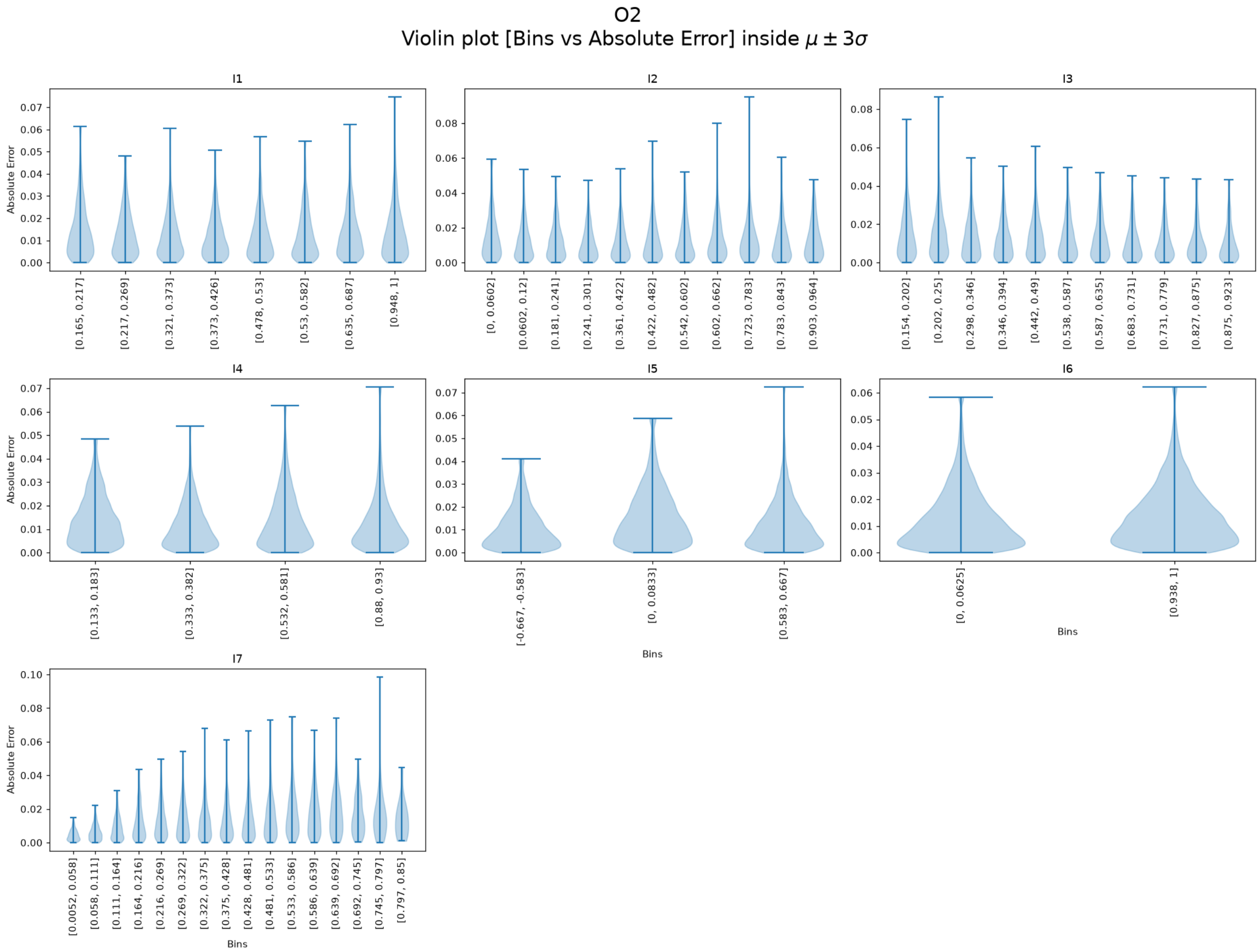

O2
Violin plot [Bins vs Residue] inside $\mu \pm 3\sigma$
I1
I2
I3
I4
I5
I6
I7
Residue
Bins
[0.165, 0.217] [0.217, 0.269] [0.321, 0.373] [0.373, 0.426] [0.478, 0.53] [0.53, 0.582] [0.635, 0.687] [0.948, 1]
[0, 0.0602] [0.0602, 0.12] [0.181, 0.241] [0.241, 0.301] [0.361, 0.422] [0.422, 0.482] [0.542, 0.602] [0.602, 0.662] [0.723, 0.783] [0.783, 0.843] [0.903, 0.964]
[0.154, 0.202] [0.202, 0.25] [0.298, 0.346] [0.346, 0.394] [0.442, 0.49] [0.538, 0.587] [0.587, 0.635] [0.683, 0.731] [0.731, 0.779] [0.827, 0.875] [0.875, 0.923]
[0.133, 0.183] [0.333, 0.382] [0.532, 0.581] [0.88, 0.93]
[-0.667, -0.583] [0, 0.0833] [0.583, 0.667]
[0, 0.0625] [0.938, 1]
[0.0052, 0.058] [0.058, 0.111] [0.111, 0.164] [0.164, 0.216] [0.216, 0.269] [0.269, 0.322] [0.322, 0.375] [0.375, 0.428] [0.428, 0.481] [0.481, 0.533] [0.533, 0.586] [0.586, 0.639] [0.639, 0.692] [0.692, 0.745] [0.745, 0.797] [0.797, 0.85]

Absolute Error Violin Plots per Output (binned by input)

Absolute Error — O1



Absolute Error — O2

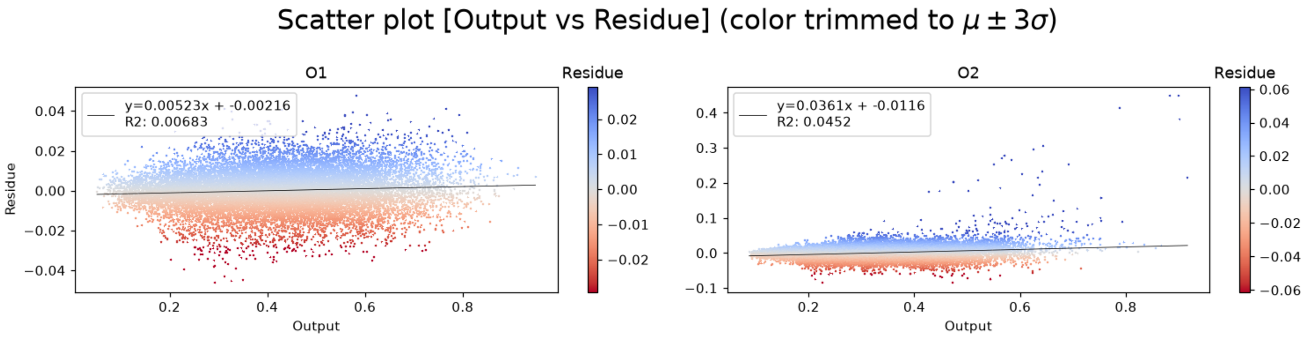


P(E|Y) — Error Conditional on Outputs

We study whether the error depends on the output magnitude. A significant trend (Pearson or Spearman $p < 0.05$) means the model is more accurate in some output ranges than others — a form of heteroscedasticity.

Residue vs True Output — Scatter

The legend shows the least-squares fit and its R^2.



Residue Trend Tests (Pearson / Spearman)

Each output's residue against its own true value, as the extended validation notebook computes them: correlation coefficient, trend-test p-value ($p < 0.05$ = significant trend) and, for Pearson, the normalized slope. The full cross-output tables are in the extended validation HTML.

	O1	O2
Pearson coeff.	0.0827	0.2126
p-value	1.79e-28	7.81e-182
Slope (normalized)	0.05	0.0561

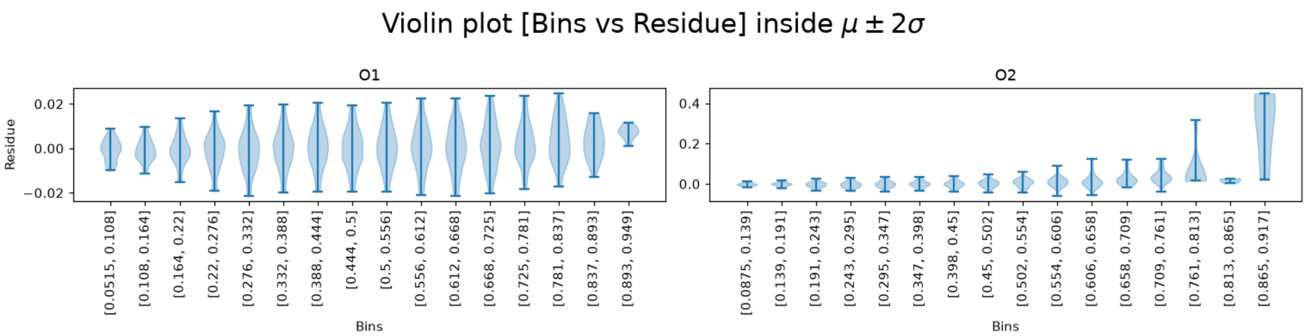
Residue vs. its true output — Pearson trend test

	O1	O2
Spearman coeff.	0.0759	0.1545
p-value	3.07e-24	6.05e-96

Residue vs. its true output — Spearman trend test

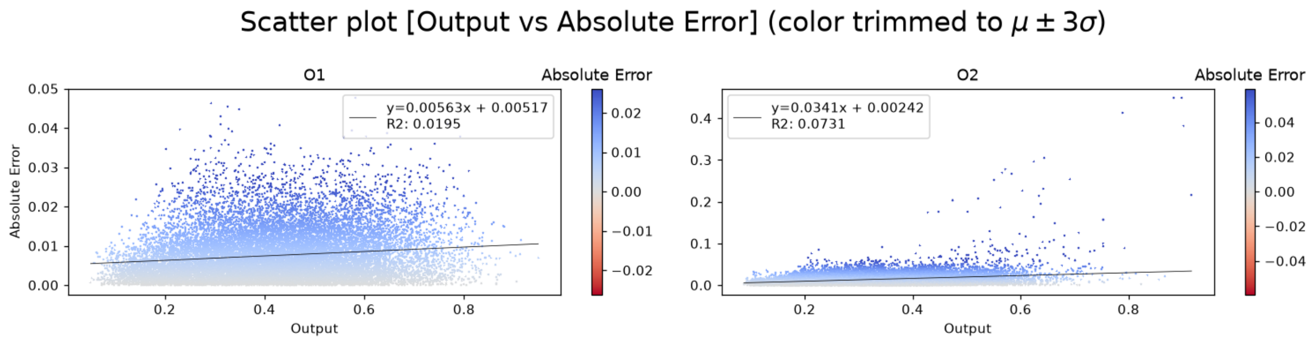
Residue Violin vs True Output Bins

Bins determined by Sturges rule. Median line dashed at 0.



Absolute Error vs True Output — Scatter

The legend shows the least-squares fit and its R^2.



Absolute Error Trend Tests (Pearson / Spearman)

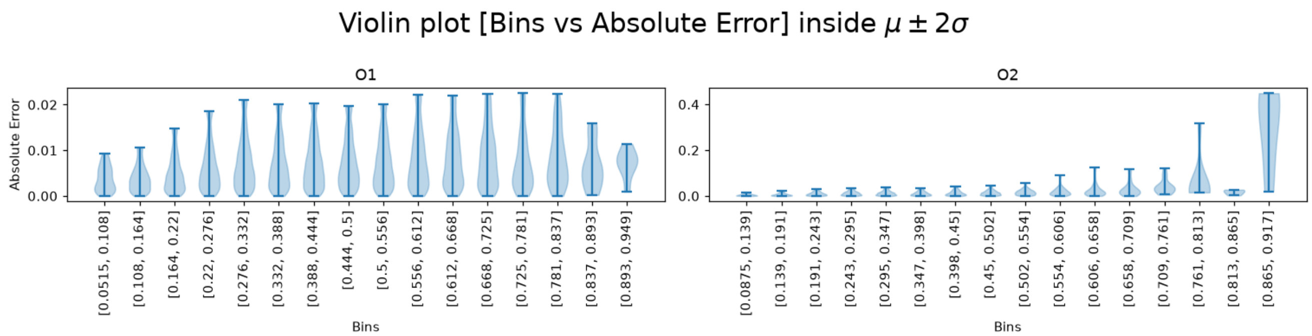
	O1	O2
Pearson coeff.	0.1398	0.2703
p-value	1.04e-78	7.57e-297
Slope (normalized)	0.106	0.063

Absolute error vs. its true output — Pearson trend test

	O1	O2
Spearman coeff.	0.1383	0.2364
p-value	4.37e-77	1.6e-225

Absolute error vs. its true output — Spearman trend test

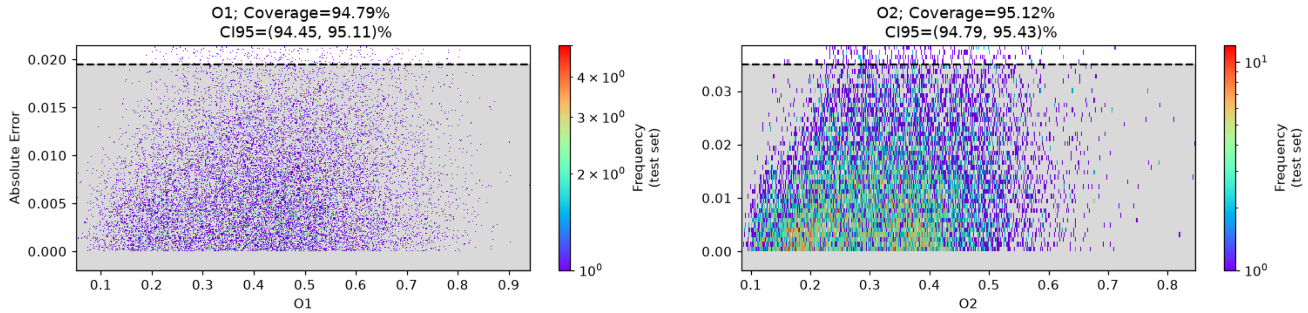
Absolute Error Violin vs True Output Bins



Uncertainty Analysis

A global binned uncertainty model (95th percentile, right-tailed) is trained on the validation split of the absolute error and evaluated on the held-out test split, as in the extended validation notebook. Coverage is the proportion of test points whose error falls inside the predicted bound; it should be close to 95%.

Coverage Plot



Coverage Table

	O1	O2	TOTAL COV
Coverage	94.79	95.12	—
CI	[94.45, 95.11]	[94.79, 95.43]	—

Empirical coverage of the 95th-percentile uncertainty bound on the test split.